

Recap

Caching aims at improving web performance by storing HTTP response messages “close” to the users for possible reuse

Three approaches to assess the validity:

- Expiration mechanisms
- Validation mechanisms
- Heuristics

Benefits of caching measured in terms of hit rate (availability + validity)

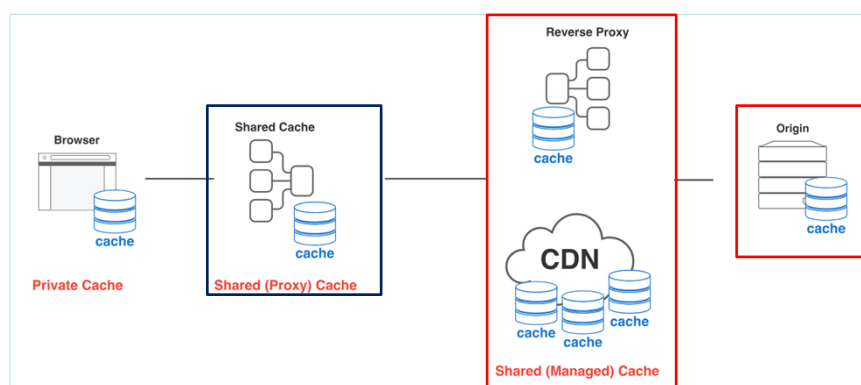
Hit rate also measures the fraction of “saved” bytes

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Managed caches



Managed caches are deployed by service providers to offload the origin servers and deliver content efficiently (web accelerators)

- Reverse proxies
- Content Delivery Networks (CDNs)

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Forward proxy

A single point of access and control located at the client side

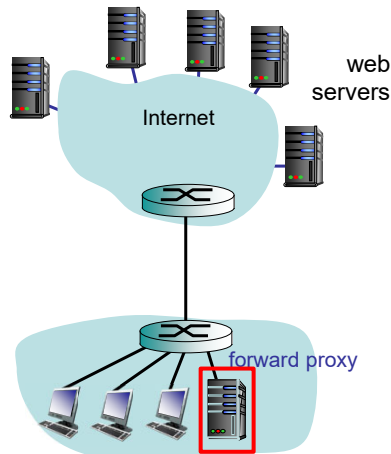
Several functionalities: web caching, access to Internet of users behind firewalls, filter/block traffic

Forward proxies hide client identities

Closed/private not transparent to users

Open proxies available for general public access and proxies left open due to misconfigurations

Open proxies are often abused for various types of attacks (anonymity)



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Reverse proxies

Reverse proxies are known as gateways or server accelerators and represent a single point of access and control at the server side

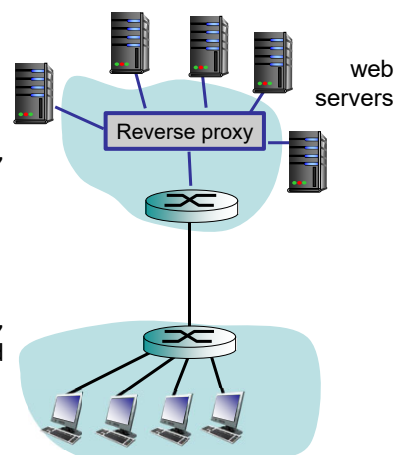
Multiple web servers share a single URL space

Reverse proxies protect web server identity, location and structure of the local network

Completely transparent for clients

Used for caching/prefecthing (static content), balance the load across multiple servers and access content located behind firewalls

Good defense against security attacks, improvements of performance and availability



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Content Delivery Networks

Content Delivery Networks represent an additional layer between clients and web servers introduced to speedup the access to web resources

They consist of a (typically big or very big) set of edge servers geographically distributed over multiple networks (i.e., Points of Presence)

The idea is to replicate web content across groups of edge servers “close” to the clients

Webmasters (together with CDN providers) decide the content distribution policies in terms of “what” and “where” (pull model)

Solution completely transparent to users

Edge servers might share hostnames

Client requests are routed to the topologically closest or “best” edge server (e.g., IP anycast)

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Examples

www.abc.com.	300	IN	CNAME	d2iww1xxkqpmiz.cloudfront.net.
d2iww1xxkqpmiz.cloudfront.net.	60	IN	A	108.157.188.77
d2iww1xxkqpmiz.cloudfront.net.	60	IN	A	108.157.188.76
d2iww1xxkqpmiz.cloudfront.net.	60	IN	A	108.157.188.23
d2iww1xxkqpmiz.cloudfront.net.	60	IN	A	108.157.188.40

www.sony.jp.	300	IN	CNAME	www.sony.jp.edgesuite.net.
www.sony.jp.edgesuite.net.	21183	IN	CNAME	a488.r.akamai.net.
a488.r.akamai.net.	20	IN	A	23.44.51.106
a488.r.akamai.net.	20	IN	A	23.44.51.107

www.mit.edu.	1800	IN	CNAME	www.mit.edu.edgekey.net.
www.mit.edu.edgekey.net.	60	IN	CNAME	e9566.dscb.akamaiedge.net.
e9566.dscb.akamaiedge.net.	20	IN	A	23.50.153.111

www.corriere.it.	238	IN	CNAME	cdn.corriere.it.
cdn.corriere.it.	238	IN	CNAME	na-eu-corriere.map.fastly.net.
na-eu-corriere.map.fastly.net.	45	IN	A	151.101.241.50

www.garmin.com.	98	IN	CNAME	www.garmin.com.cdn.cloudflare.net.
www.garmin.com.cdn.cloudflare.net.	98	IN	A	104.17.148.16
www.garmin.com.cdn.cloudflare.net.	98	IN	A	104.17.147.16

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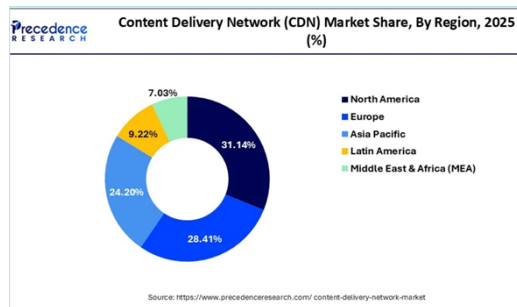
CDNs (cont'd)

Complex (and expensive) solutions mostly appropriate for web services that operate at large scale (vs localized websites)

CDNs provide many diverse benefits, e.g., improved PLT by reducing latency, reduced bandwidth usage, effective management of high peaks of requests, effective defense against security attacks (e.g., DDoS attacks), improved availability

More than half of the world Internet traffic over CDNs (30% over Akamai servers)

Proprietary technologies

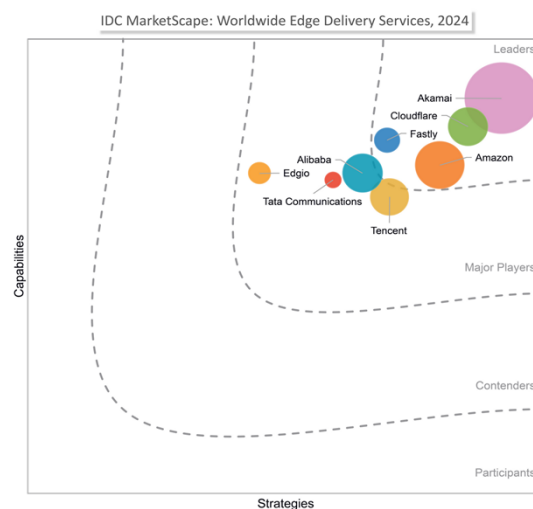


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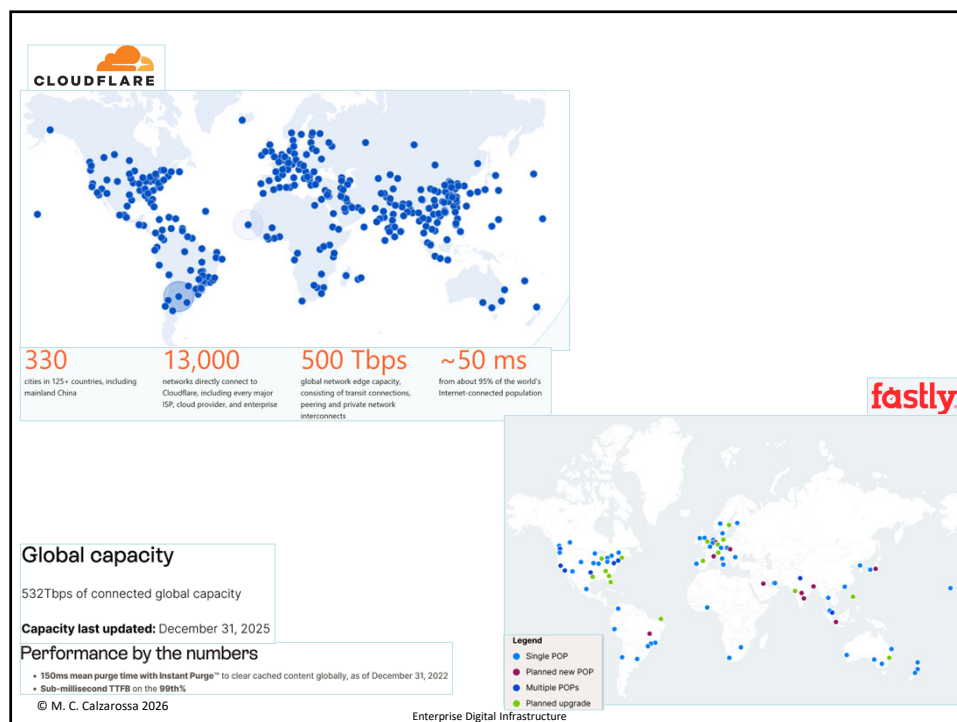
Main CDN providers



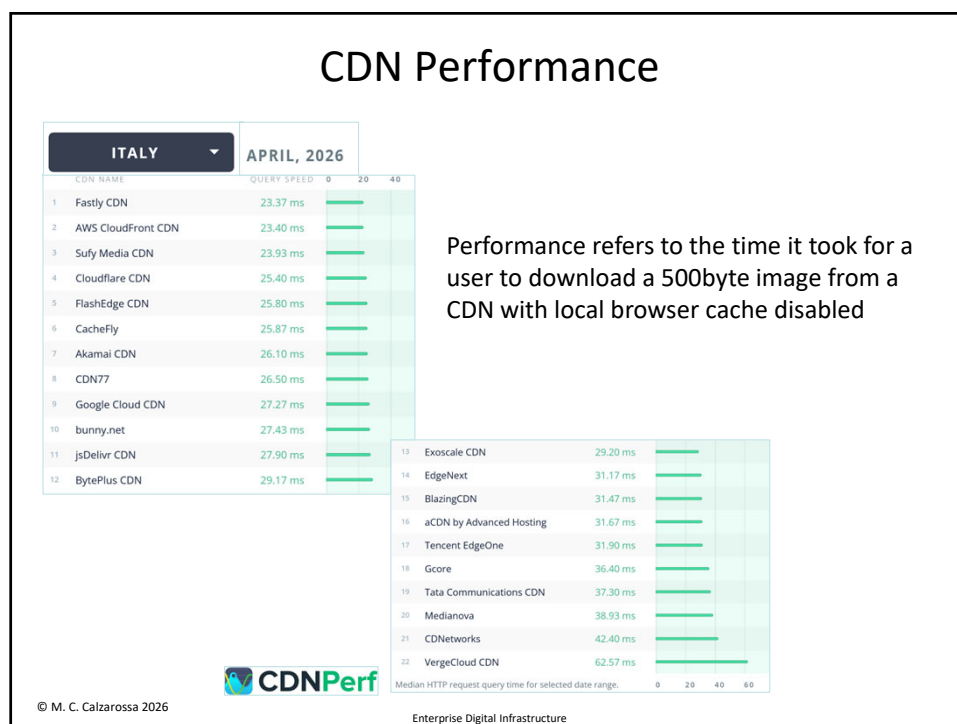
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HTTP/2

HTTP/2 is one of the members of the HTTP family

It represents the first *major revision* of the HTTP protocol that tries to overcome some limitations of HTTP/1.1 [[RFC 9113 “HTTP/2”](#) published in June 2022, first standard published in 2015]

Based on SPDY, a protocol announced by Google in 2009 and installed on its servers since 2011

The main objective of HTTP/2 is to enable a more efficient usage of network resources and reduce latency

Main changes refer to how the data is formatted and transported between clients and web servers (e.g., framing and multiplexing layers)

Completely transparent to the users

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HTTP/2 main features

- Single (persistent) TCP connection
- Binary framing
- Multiplexed requests
- Prioritized requests
- Compression
- Server Push

HTTP/2 uses by default TLS (**h2**), but it can also run over cleartext TCP (**h2c**)

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Binary framing

Binary framing is a layer that dictates how the HTTP messages are encapsulated and transferred between clients and web servers

A binary protocol is more efficient to parse, more compact on the wire and less error-prone with respect to textual protocols

Framing is a method to wrap information in a way that makes it easy for consumers of the the protocol to create, read and parse

Frame: basic protocol unit used in all exchanges between a client and a server

Stream: a series of frames making up pairs of HTTP requests/responses

Communications are performed over a single TCP connection that can carry any number of bidirectional streams

A stream is assigned a unique identifier (odd identifier for a client-initiated stream; even identifier for a server-initiated stream) and an optional priority

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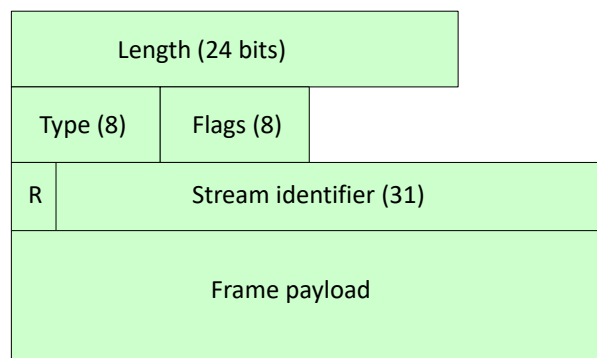
Frames

HTTP request and response messages are mapped into frames

A frame has a fixed structure (9 bytes headers) and variable length payload

Ten types of frames:

- HEADERS
- DATA
- SETTINGS
- WINDOW_UPDATE
- GOAWAY
- RST_STREAM
- PUSH_PROMISE
- PRIORITY
- PING
- CONTINUATION



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